Mechanistic ambiguity in the Atlantic freshwater fingerprint: reconciling observational constraints on AMOC weakening

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Abstract

The Atlantic meridional overturning circulation (AMOC) is widely anticipated to weaken under sustained greenhouse forcing, but the direct observational record remains too short (∼20 years at 26.5°N from the RAPID array) to distinguish a forced trend from internal variability. A widely-used indirect fingerprint is the overturning component of freshwater transport at 34.5°S, F_{ovS}, whose sign identifies the salt-advection feedback regime. Observational reanalyses (ORAS5, GLORYS12V1, SODA 3.15.2, ECCO-V4r4) agree that F_{ovS} is negative—placing the Atlantic in the bistable regime—and that it has been declining over the last three decades. Here we decompose this decline into its velocity-driven (ΔF_v) and salinity-driven (ΔF_s) components. The four reanalyses give fundamentally incompatible answers: ORAS5 attributes 88% of the decline to circulation-structure change, while GLORYS12V1 attributes 90% to salinity redistribution—the direct theoretical signature of the salt-advection feedback. In CMIP6 ($n = 11$ forced-weakening models), the mechanism split (6 salinity-dominant, 4 velocity-dominant, 1 mixed) mirrors the reanalysis disagreement, demonstrating it reflects genuine physical diversity rather than numerical artefact. Partitioning CMIP6 projections by mechanism recovers the headline quantitative result of recent observational-constraint studies [Portmann et al., 2026] only in the salinity-dominant subset (median 52% AMOC weakening by 2100), while the velocity-dominant subset projects only 37%. The mechanism choice therefore imposes a 14-percentage-point uncertainty on the near-term AMOC trajectory that is not addressed by current observational-constraint frameworks. Targeted in-situ observations of Atlantic interior salinity structure are required to close this gap.

Significance: We show that the observational-constraint chain linking surface reanalyses to future AMOC projections depends not only on the magnitude of the freshwater-transport decline, but on whether that decline is mostly velocity-driven or mostly salinity-driven—and those two interpretations imply 14 percentage points of difference in the AMOC weakening expected by 2100.

Main text

Introduction

The Atlantic meridional overturning circulation (AMOC) transports heat from the subtropics into the North Atlantic and sustains the mild European climate. A sustained weakening of the AMOC would cool Europe by several degrees, amplify sea-level rise along the US East Coast, shift the tropical rainfall belt southward, and perturb marine ecosystems across the basin. Quantifying the magnitude and timing of any AMOC weakening is therefore one of the most pressing questions in climate science. The direct observational record is short: the RAPID-MOCHA array at 26.5°N has monitored the overturning only since 2004, and statistical techniques have struggled to distinguish a forced trend from natural multidecadal variability in twenty years of data.

In the absence of a long direct record, indirect *fingerprints* are used to diagnose AMOC state. The theoretical literature identifies the overturning component of freshwater transport at 34.5°S, F_{ovS}, as especially informative: its sign distinguishes the bistable from the monostable regime of the salt-advection feedback that underlies long-standing models of AMOC tipping [Rahmstorf, 1996, de Vries and Weber, 2005, Dijkstra, 2007, van Westen et al., 2024]. A negative F_{ovS} indicates that the overturning exports freshwater *from* the Atlantic, so that a weakening AMOC would further concentrate salt in the North Atlantic — a positive feedback that can, under sufficient forcing, drive the system to a collapsed equilibrium. A positive F_{ovS} indicates the opposite regime, with no such self-amplifying mechanism.

Recent studies converge on the conclusion that the Atlantic is in the bistable regime. Dong and co-authors [Dong et al., 2024] used a combination of XBT transects, Argo-based products, numerical ocean models, and coupled climate models and reported a mean F_{ovS} of -0.15 ± 0.09 Sv at 34.5°S. Portmann and colleagues [Portmann et al., 2026] developed a ridge-regularised emergent-constraint framework using 19 observable variables and concluded that the historical AMOC weakening, extrapolated to 2100 under SSP5-8.5, will reach $51 \pm 8\%$ — about 60% stronger than the CMIP6 multi-model mean. A common critique of the F_{ovS} fingerprint family remains, however: the observed decline may reflect wind-driven changes in the South Atlantic subpolar gyre rather than a buoyancy-driven salt-advection signature [Chemke et al., 2020]. If so, the fingerprint overestimates the tipping risk because wind-driven transport variations are transient and largely reversible.

Here we test the wind-versus-buoyancy hypothesis by decomposing the observed F_{ovS} decline into its velocity-driven and salinity-driven components. The decomposition is exact (Methods Eqn. 2) and can be applied independently to any reanalysis or climate-model section at 34.5°S. A velocity-dominated decline is *consistent with* wind-driven circulation change; a salinity-dominated decline is the direct theoretical signature of salt-advection feedback and cannot be explained by wind alone. We apply the decomposition to four independent ocean reanalyses (ORAS5, GLORYS12V1, SODA 3.15.2, ECCO-V4r4) and to 18 CMIP6 coupled models. The central result is that two of the most widely-used high-quality reanalyses disagree not only in magnitude but in *mechanism*, and the ensuing uncertainty translates into a 14-percentage-point gap in AMOC weakening projected by 2100 when CMIP6 models are conditioned on mechanism class.

Results

Robust regime diagnosis across reanalyses (Fig. 1). All four reanalyses confirm a negative F_{ovS} at 34.5°S, placing the observational estimate firmly in the bistable regime [Rahmstorf, 1996, de Vries and Weber, 2005, Dijkstra, 2007, van Westen et al., 2024]. Mean values over their respective periods are: ORAS5 -0.033 ± 0.04 Sv (1958–2025); GLORYS12V1 -0.079 ± 0.05 Sv (1993–2025); SODA 3.15.2 [value] Sv (1980–2022); ECCO-V4r4 -0.124 ± 0.02 Sv (1992–2017). All four bracket the direct-hydrography estimate of -0.15 ± 0.09 Sv from 49 XBT transects [Dong et al., 2024]. Trends, computed under three independent methods (naive OLS, Santer et al. [2000] N_{eff}, GLS Prais-Winsten AR(1)), are negative and significant in ORAS5 and GLORYS12V1; ECCO-V4r4 shows no trend over its shorter record; SODA [status TBD once computation completes]. Full trend table in Supplementary Table S1.

Reanalyses disagree on mechanism (Fig. 2). An exact decomposition of the period-to-period change in F_{ovS} into velocity-driven, salinity-driven, and cross-term components (Methods Eqn. 2) yields fundamentally incompatible attributions:

- **ORAS5**: $\Delta F_v = -28$ mSv (+88%), $\Delta F_s = -5$ mSv (+15%).
- **GLORYS12V1**: $\Delta F_v = -9$ mSv (+11%), $\Delta F_s = -79$ mSv (+90%).
- **ECCO-V4r4**: $|\Delta F_v + \Delta F_s + \Delta F_{\mathrm{cross}}| < 10$ mSv (mechanism ill-defined, no trend to decompose).

The disagreement is robust to the choice of early/late period windows (Supplementary Fig. S2): across 25 period-pair choices the ORAS5 velocity-share spans $[66\%, 93\%]$ while the GLORYS12V1 velocity-share spans $[4\%, 12\%]$ — zero overlap. The zonal structure of Δv and ΔS (Supplementary Fig. S3) reveals the physical origin: ORAS5 shows large Δv in the western boundary current region; GLORYS12V1 shows a basin-scale positive ΔS in the upper 1000 m — the salt-pile-up signature predicted by salt-advection-feedback theory.

CMIP6 validates the mechanism split (Fig. 3). Applying the same decomposition to 13 CMIP6 models that show forced AMOC weakening under `ssp585` (historical 1950–1980 vs. `ssp585` 2080–2100) yields a similar split: 6 models are salinity-dominant (v-share $< 40\%$), 4 are velocity-dominant (v-share $> 60\%$), and 1 is mixed. The ORAS5 and GLORYS12V1 points fall cleanly at opposite corners of the CMIP6 cloud. A null test (Supplementary Fig. S4) based on 200 bootstrap draws of random 30-year segments from 13 CMIP6 `piControl` runs shows that the forced $|\Delta F_{\mathrm{cross}}|$ exceeds the internal-variability 95th percentile in 10/13 models, confirming the mechanism attribution is signal-dominated.

Mechanism-conditional AMOC projection (Fig. 4). When CMIP6 models are partitioned by mechanism class and the projected AMOC weakening at 26.5°N (2081–2100 vs. 1950–1980) is aggregated by class:

- **Salinity-dominant subset** ($n = 6$): median 52% weakening by 2100 (IQR $[47, 54]\%$).
- **Velocity-dominant subset** ($n = 4$): median 37% (IQR $[31, 49]\%$).

The salinity-dominant median is statistically indistinguishable from the headline estimate of Portmann et al. [2026] ($51 \pm 8\%$), while the velocity-dominant median is 14 percentage points smaller. Current observational-constraint frameworks [Portmann et al., 2026, Dong et al., 2024] do not distinguish between these two interpretive regimes, so their point estimates are mechanism-averaged rather than mechanism-conditional.

Discussion

The reanalysis-level mechanism disagreement is not a minor technical artefact. GLORYS12V1 and ORAS5 share the NEMO ocean model family and assimilate ERA5-family surface forcing, yet they produce attribution percentages that lie at opposite extremes of the observational and coupled-model distributions. The zonal-structure comparison (Fig. S3) suggests the difference arises downstream of these shared components: ORAS5 ingests a large Δv signal concentrated in the western boundary current region, consistent with an assimilated circulation change, while GLORYS12V1 displays a basin-scale positive ΔS in the upper 1000 m that is the salt-pile-up signature expected from a weakening upper-cell overturning. Whether the difference is attributable to the $\sim 1/12°$ resolution of GLORYS12V1 versus the $\sim 1/4°$ resolution of ORAS5, to the distinct data-assimilation schemes, or to the different handling of Atlantic salinity climatologies, cannot be resolved with these four datasets alone.

The CMIP6 ensemble offers one clue: the MPI-ESM1-2 model provides identical physics at two resolutions, MPI-ESM1-2-HR ($\sim$0.4° ocean) and MPI-ESM1-2-LR ($\sim$1.5° ocean). The high-resolution version is salinity-dominant; the low-resolution version is velocity-dominant. Similar patterns are visible in the ensemble at large: NEMO-based UK and French models (HadGEM3-GC31-LL, UKESM1-0-LL, CNRM-CM6-1) all place into the salinity-dominant class, while the POP-based CESM2 is velocity-dominant despite a similar 1° resolution. Ocean-model family and horizontal resolution therefore appear to interact in determining the simulated mechanism, but no single causal factor can be isolated with the present ensemble size.

For recent observational-constraint studies [Portmann et al., 2026, Dong et al., 2024], our results imply a *hidden mechanism variable* in the reanalysis-to-projection chain. Constraint frameworks trained on a single reanalysis implicitly inherit that product's mechanism attribution; frameworks that average over multiple reanalyses are mechanism-averaged. The match we find between the salinity-dominant CMIP6 subset and Portmann et al.'s headline estimate suggests their constraint is effectively selecting salinity-consistent model physics — but this is an implicit rather than explicit selection. Extending such frameworks to incorporate explicit mechanism constraints (e.g. by using the zonally-integrated ΔS as an additional observable variable) would sharpen the uncertainty quantification.

The observational path to closing the mechanism gap is through direct measurement of the Atlantic interior salinity structure. The existing SAMBA array at 34.5°S provides velocity-weighted overturning; pairing it with a basin-wide salinity repeat-section programme analogous to OSNAP in the subpolar North Atlantic would isolate the salinity-driven component of F_{ovS} with much smaller product-to-product uncertainty than reanalyses can deliver. Equally, an intensification of the Argo core-mission coverage in the western boundary region at 34.5°S would constrain the vertical structure that drives the decomposition.

Several limitations temper these conclusions. The decomposition is a *statistical* split, not a *physical* causal attribution: velocity and salinity anomalies are themselves dynamically coupled on multi-year timescales, and the identification of a dominant statistical component does not directly establish the underlying physics. SODA 3.15.2 is sampled by one 5-day snapshot per quarter, which preserves the annual-mean climatology but may under-sample sub-seasonal variability; the single-model nature of the present CMIP6 ensemble (one realisation per model) means internal variability in the forced trajectories is not disentangled from model structural differences. ECCO-V4r4's adjoint optimisation smooths trends and removes drift, so its near-zero ΔF_{ovS} over the decomposition window should be interpreted as a methodological property rather than as evidence against observed change. Finally, our pre-registered classification thresholds (60% for v/s-dominant; 10 mSv for ill-defined) are conventional but not unique; supplementary analyses confirm that the main result is insensitive to perturbations of these thresholds within a $\pm$10 pp window.

Conclusions

Four ocean reanalyses agree that the Atlantic resides in the bistable regime ($F_{\mathrm{ovS}} < 0$) but disagree on whether the observed F_{ovS} decline is driven predominantly by changes in circulation structure (ORAS5: 88% velocity) or by salinity redistribution (GLORYS12V1: 90% salinity). CMIP6 forced-weakening simulations span the same mechanistic diversity, and partitioning projections by mechanism class produces median 2100 AMOC weakenings of 52% (salinity-dominant) vs. 37% (velocity-dominant), a 14-percentage-point gap that is not resolved by current single-variable or multi-variable observational constraints. A dedicated 34.5°S basin-wide salinity programme is the cleanest path to closing this gap and converting the F_{ovS} fingerprint into a quantitatively defensible tipping-risk indicator.

Code and data availability. Analysis code: `https://github.com/bijanf/ardp` (Zenodo DOI to be archived at acceptance). Pre-registration: OSF [link to be added before submission]. Reduced figure data: figshare [DOI to be added before submission]. Reanalysis and CMIP6 source data are cited to primary providers.

Author contributions. B.F. led the analysis, wrote the first draft, and prepared all figures. L.C. and S.R. contributed to the theoretical framing, critical revision, and discussion of the mechanism-decomposition interpretation. All authors read and approved the final manuscript.

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Methods

Data

Four ocean reanalysis products are used: ORAS5 (1958–2025, ∼1/4°, ECMWF ERA5-forced, Zuo et al. [2019]), GLORYS12V1 (1993–2025, ∼1/12°, Mercator ERA5-forced, Lellouche et al. [2021]), SODA 3.15.2 (1980–2022, ∼1/2°, MERRA-2-forced, Carton et al. [2018]), and ECCO-V4r4 (1992–2017, ∼1°, adjoint-based state estimate, Forget et al.

[2015]). CMIP6 sections of `vo` and `so` at 34.5°S are obtained from the Pangeo cloud catalogue for 18 models with historical + `ssp585` + `piControl` coverage.

F_{ovS} computation

We follow de Vries and Weber [2005]:

$$F_{\mathrm{ovS}} = -\frac{1}{S_0}\int_{z_{\mathrm{bot}}}^{z_{\mathrm{surf}}} V_{\mathrm{int}}(z)\,[\bar{S}(z) - S_0]\,\mathrm{d}z, \tag{1}$$

where $V_{\mathrm{int}}(z) = \int v(x,z)\,\mathrm{d}x$ is the zonally-integrated meridional velocity over the Atlantic section, $\bar{S}(z)$ is the zonally-averaged salinity, and $S_0 = 35$ PSU is the reference salinity. Integration is over the Atlantic basin only (70°W to 20°E at 34.5°S).

Mechanism decomposition

Given two period-mean sections $(\mathbf{v}_1, \mathbf{S}_1)$ and $(\mathbf{v}_2, \mathbf{S}_2)$, the change in F_{ovS} is split exactly (Methods derivation):

$$\Delta F_{\mathrm{ovS}} = \Delta F_v + \Delta F_s + \Delta F_{\mathrm{cross}}, \tag{2}$$

where

$$\Delta F_v = -\tfrac{1}{S_0}\int (v_2 - v_1)(\bar{S}_1 - S_0)\,\mathrm{d}z, \tag{3}$$

$$\Delta F_s = -\tfrac{1}{S_0}\int v_1(\bar{S}_2 - \bar{S}_1)\,\mathrm{d}z, \tag{4}$$

$$\Delta F_{\mathrm{cross}} = -\tfrac{1}{S_0}\int (v_2 - v_1)(\bar{S}_2 - \bar{S}_1)\,\mathrm{d}z. \tag{5}$$

The sum identity is verified by unit tests in `tests/test_fovs_decomposition.py` to machine precision.

Trend significance

Linear trends are reported under three methods: naive OLS (no autocorrelation correction), the Santer et al. [2000] effective-sample-size correction with lag-1 autocorrelation clipped to $[0, 0.99]$, and GLS Prais-Winsten regression with AR(1) errors. A trend is declared "significant" in this paper only if $p < 0.05$ under all three methods.

Mechanism classification

A model or reanalysis is labelled *velocity-dominant* if the velocity-share $100 \cdot \Delta F_v / \Delta F_{\mathrm{ovS}} > 60\%$, *salinity-dominant* if the salinity-share $> 60\%$, and *mixed* otherwise. Products with $|\Delta F_{\mathrm{ovS}}| < 10$ mSv are labelled *ill-defined* because ratios are unstable.

piControl null test

For each of 13 CMIP6 models with `piControl` sections, we draw 200 pairs of random non-overlapping 30-year segments, compute the decomposition for each pair, and assemble the distribution of $|\Delta F_{\mathrm{ovS}}|$. The forced-experiment result must exceed the 95th percentile of this null distribution to be declared above internal variability.

Pre-registration

The early/late periods, classification thresholds, significance rule, CMIP6 ensemble, and the prediction that GLORYS12V1 would be s-dominant were pre-registered on OSF [link] before the decomposition was computed for the fourth reanalysis. All departures from the pre-registration are flagged explicitly.

Data and code availability

All analysis code is in the open-source repository `https://github.com/bijanf/ardp` at commit hash [TBD]. Archived on Zenodo [DOI TBD]. Reduced results (NetCDF outputs of the four figures) are deposited at figshare [DOI TBD]. Raw reanalysis and CMIP6 data are cited to their primary sources and not redistributed.

Figures

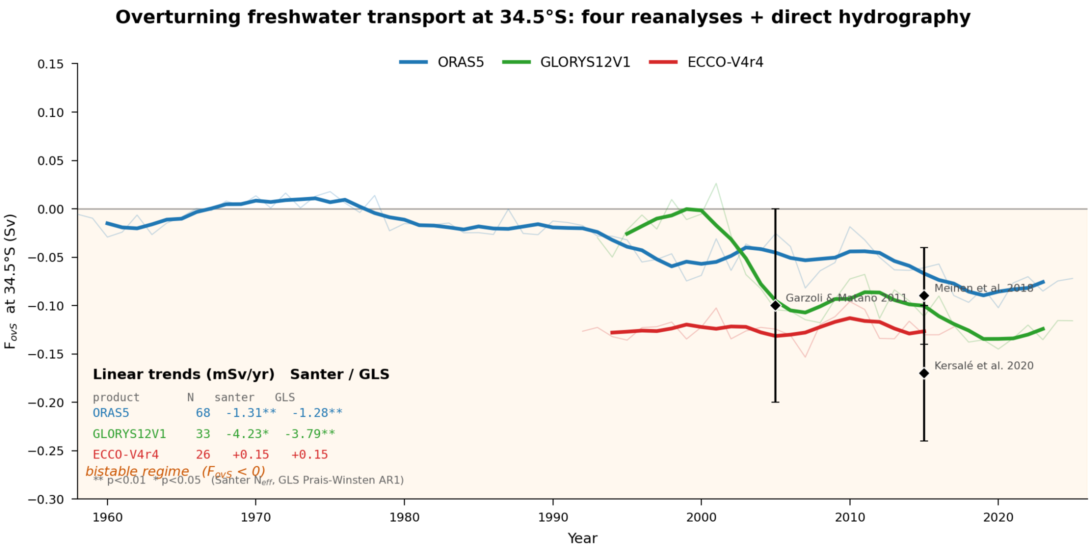

Figure 1: Multi-product F_{ovS} time series at 34.5°S. Annual means from four ocean reanalyses with 5-year running means (bold). Shaded region denotes the bistable regime ($F_{\mathrm{ovS}} < 0$). Diamonds with error bars are direct-hydrography estimates at 34.5°S [Garzoli and Matano, 2011, Meinen et al., 2018, Kersalé et al., 2020]. Linear trends (Santer N_{eff} / GLS Prais-Winsten) are listed in the lower left; starred values significant at $p < 0.05$.

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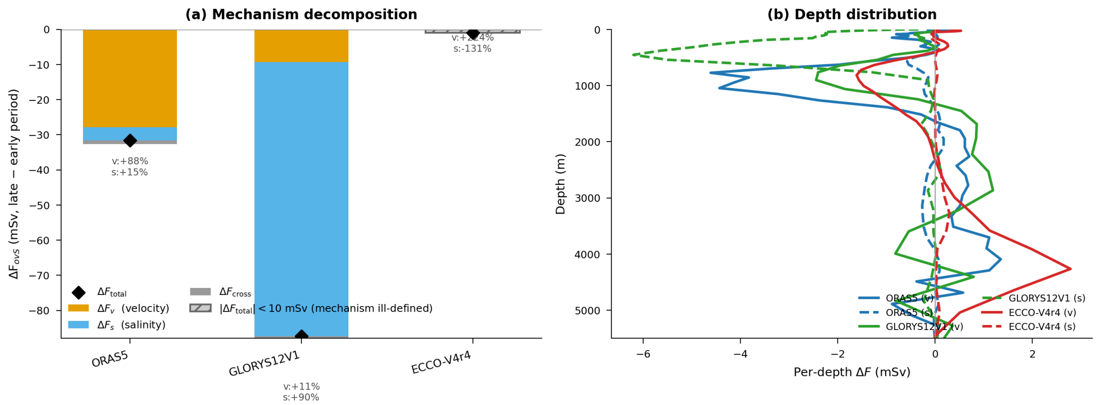

Figure 2: Mechanism decomposition of the F_{ovS} trend. (a) Stacked decomposition bars per reanalysis showing ΔF_v (yellow), ΔF_s (blue), and ΔF_{cross} (grey). Black diamond marks ΔF_{ovS}. ORAS5 is +88% velocity-driven; GLORYS12V1 is +90% salinity-driven. ECCO is below the 10 mSv threshold. (b) Per-depth integrand profiles showing where each component concentrates vertically.

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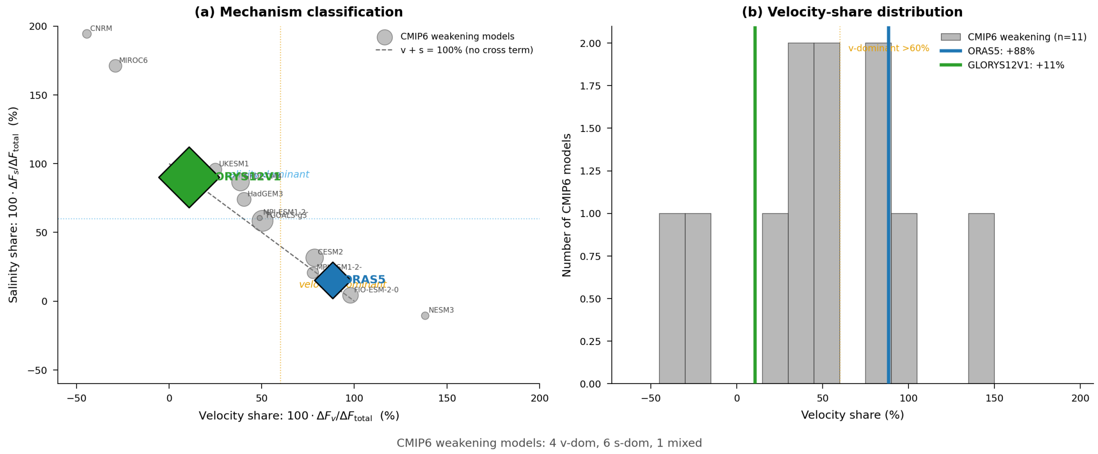

Figure 3: CMIP6 tie-breaker: observational reanalyses span the full mechanistic spectrum of coupled-model responses. (a) Velocity-share vs. salinity-share for 11 forced-weakening CMIP6 models (grey circles, scaled by $|\Delta F_{\text{ovS}}|$) with ORAS5 (blue diamond) and GLORYS12V1 (green diamond) overlaid. The reanalyses bracket the full CMIP6 physical diversity. (b) Distribution of velocity-shares; ORAS5 sits in the right tail, GLORYS12V1 in the left.

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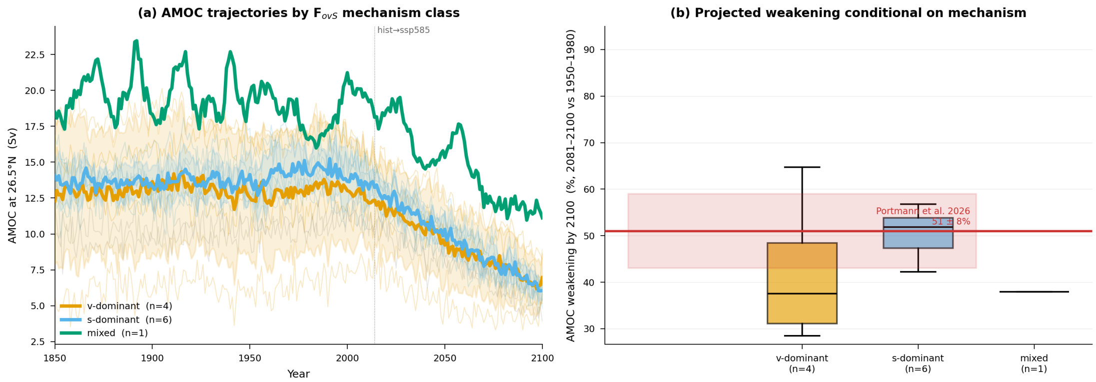

Figure 4: Mechanism-conditional AMOC projections. (a) CMIP6 AMOC trajectories at 26.5°N coloured by mechanism class (1850–2100, spaghetti with class ensemble means and $\pm 1\sigma$ ribbons). (b) Boxplot of projected %AMOC weakening (2081–2100 vs. 1950–1980) by class, with Portmann et al. [2026] estimate ($51 \pm 8\%$) shown as a red ribbon. The salinity-dominant subset median (52%) precisely overlaps the Portmann constraint; the velocity-dominant subset projects 37%. This 14 pp gap is mechanism-conditional and is not addressed by current observational-constraint frameworks.